

IOT-IMPLEMENTED ENVIRONMENTAL HYDROELECTRIC SHOWER MODULE

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IOT-IMPLEMENTED ENVIRONMENTAL HYDROELECTRIC SHOWER MODULE

Abstract

With the water's kinetic energy in turning propellers, our team wants to produce a module that could be attached to a "shower" to create some amount of electricity that can have a versatile source. The water would enter the device's tube. Then, turn the propellers in the tube, turning the axis and the gear attached to them. The gear would spin and turn gears and an axis connected to a generator underneath the tube. From the generator, electricity would be produced; using this electricity, we plan to create an IoT system(including Bluetooth, LCD, and Arduino) to effectively control an automatic valve that stops or lets water flow. Overall, it will provide a shower part that encourages using renewable energy and saving water, becoming a new solution to electricity overconsumption. Furthermore, we attempt to research types of propellers, types of gears, and several designs that would effectively create electricity environmentally.

Keywords: Hydroelectricity, Propeller, Gear, IoT, Water Conservation

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I. INTRODUCTION

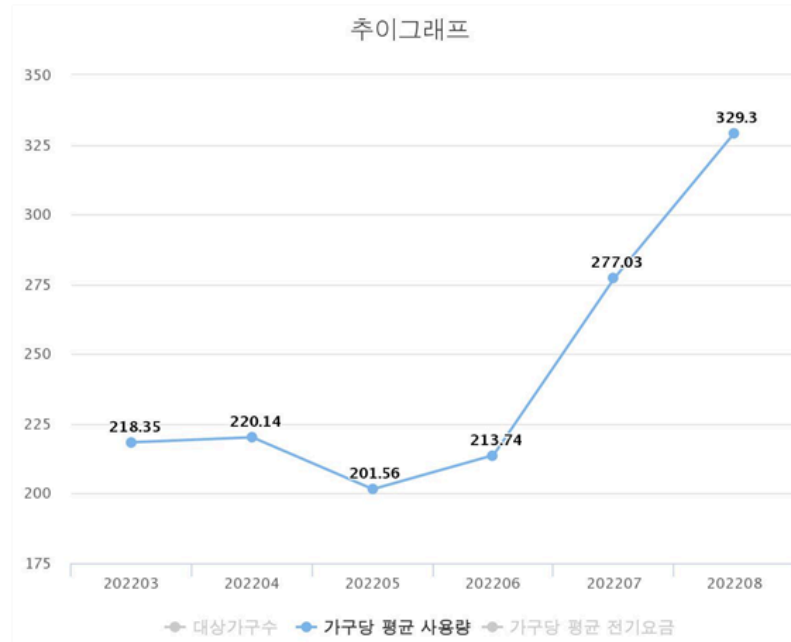
A. REASON/HYPOTHESIS

When I turned on the shower, I saw the water dripping and I thought I could create energy with that movement. This machine was created to answer the question of whether the kinetic energy of falling water could be used. First of all, how much electricity does a typical household use? There are various electronic products in a regular house, such as refrigerators, TVs, and chargers. Most households consume 300 ~ 500 kWh, but generate little or no electricity themselves. The method of generating electricity using sunlight or solar power is the most prevalent, but even this is produced in a small amount. In addition to using solar power, our team hypothesized that electricity could be made with running water from a faucet or shower through a micro-hydroelectric module.

B. IMPORTANCE

Currently, Korea consumes a large amount of electricity, ranking third in the world in per capita electricity usage. Because of the large amount of power consumed, Korea is also making efforts to actively utilize new and renewable energy such as solar heat to reduce power production and consumption costs. Among them, the most used solar panels in Korea produce 25kWh (300~500W) per one: an amount that can be used for one refrigerator. This is a very small amount compared to the average total electricity use per household (300-500 kWh). Electricity consumption is increasing day by day, but the reality is that the current technology for new and renewable energy is still insufficient to handle it. If these problems pursue, not only will electrical bills will cost more, but carbon emissions will increase rapidly, exacerbating environmental destruction. To solve this, I think that many alternative methods using new and renewable energy should be applied to furniture should emerge.

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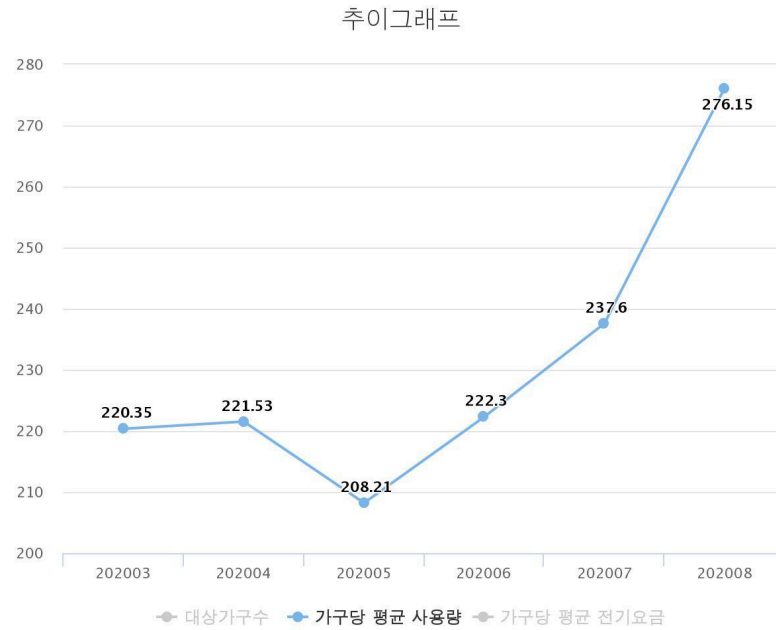


[Picture 1] Monthly Electrical Use Per Capita (kWh), 2022.03 ~ 2022.08



[Picture 2] Monthly Electric Bill Per Capita(Won), 2022

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[Picture 3] Monthly Electrical Use Per Capita (kWh), 2020.03 ~ 2020.08



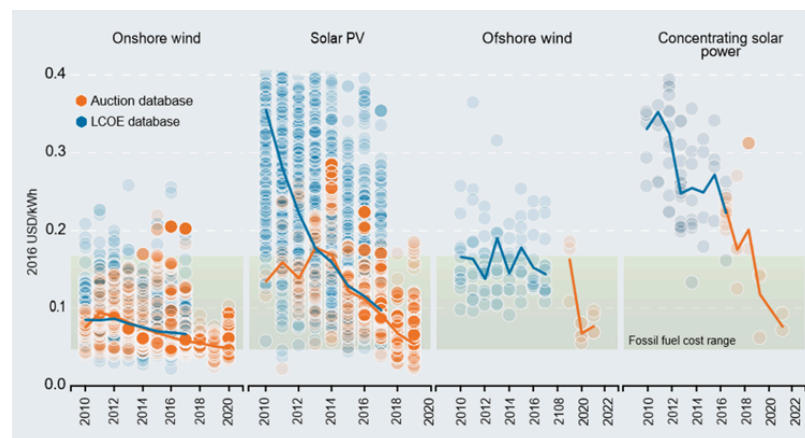
[Picture 4] Monthly Electric Bill Per Capita(Won), 2020.03 ~ 2020.08

Based on the first graph, it can be inferred that electrical consumption is greatly increasing, with the minimum consumption of 201.56 kWh and a very increased maximum usage of 329.3 kWh. Intertwined with the

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first graph, the second graph of electrical bills has a very close correlation. In 2022 May, the electrical bill is 21,126 Won, whereas in 2022 August, the bill is 40,713 Won. Both graphs show the same escalating pattern: the data initially start to decrease minimally but then steps up from May. When looking at the years 2022 and 2020, the use of electricity has swelled—explicitly shown in Augusts. In 2020 August, it was originally 276.15 kWh with a price of 29,781 Won; however, it was added by 53.15 kWh and 10,932 Won in 2022. All in all, not only does electrical use steep up in summer or winter, overall consumption is increasing annually: it is an emergency.

There is a prejudice that general hydroelectric power generation is only possible with special places and facilities such as dams and rivers. However, micro-hydropower, which has recently attracted awareness, is not only environmentally friendly and clean energy but also simple to install. It can be applied to places with low pressure and flow. Due to these advantages, it will now be possible to sufficiently generate hydroelectric power even in places where generation was impossible in the past. For example, it is now possible to generate electricity using water flow in various large and small places, such as sewage or water supply pipes flowing into septic tanks, water pipes in buildings and apartments, and faucets in homes. In other words, we can easily produce electricity anywhere around us. In this study, we conceived of small hydropower generation ideas that can be applied to showers in homes. We predict that by installing small hydropower plants, the energy used per household can be greatly reduced together with solar power. Our team also conjectured that even though this module will not be able to replace solar power, it will act as a great addition. When put on showers or anywhere water flows, we hypothesized it would reach a minimum generation of 5 V/W.



[Figure 5] 2010-2022 Generation Costs and Monthly Weighted Averages for Renewable Energy Projects

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II. BODY

A. RESEARCH GOAL

The maximum flow rate of the shower head we use every day is 12 L per minute: Liter (unit of volume)
Exceeds L (Liter).

If it is assumed that each person showers for 20 minutes each day in a household of 4 people:

$$E = 12 \cdot t \cdot p$$

E = amount of energy used per day (L)

t = amount of time taken per person (min)

p = number of people

960 L is used per person. If it is more than a month:

$$T = E \cdot d$$

T = total energy consumption at any day (L)

d = number of days

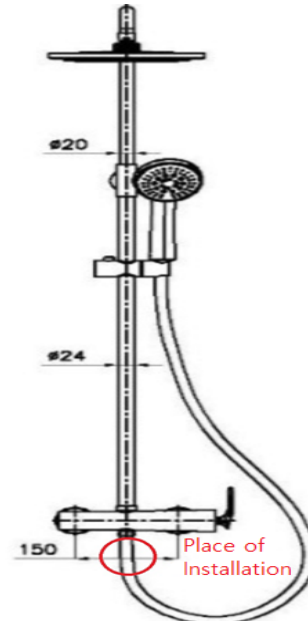
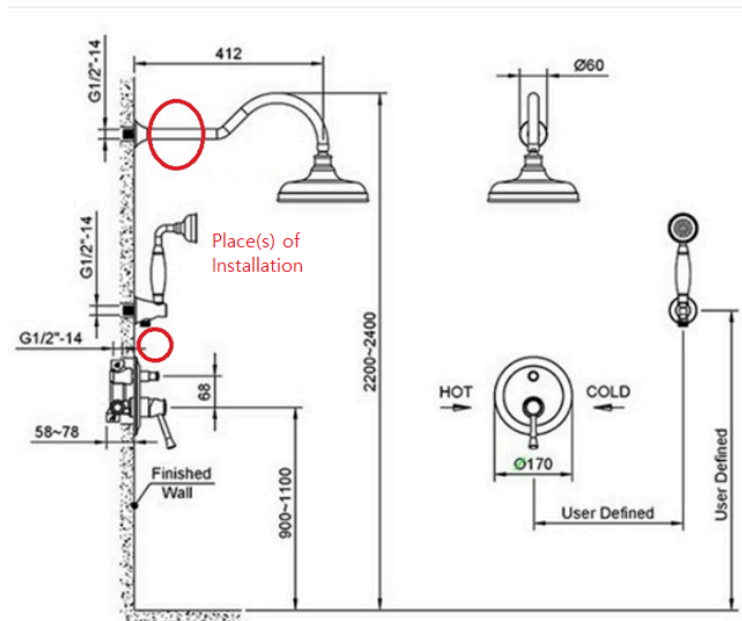
It will be approximately 28,800 L in one house.

If a great amount of water is used, the electricity produced by the micro-hydroelectric device will be substantial. Not only that, but the flow of large amounts of water can generate energy to charge the device and install motor valves and sensors. Together with this valve, ultrasonic sensors will identify if people are present or not and automatically turn the motor to stop the flow of wasted water. So, in addition to generating more electricity, it can be of great help in saving water. In addition, since the shower head is detachable, our team thought it would be more convenient to install the small hydro energy generator there. If the kinetic energy generated through small hydropower generation turns the propeller efficiently and creates a mechanical device that stores energy(i.e. rechargeable battery), it will increase electricity production per household and ultimately reduce the cost of electricity. This will create hydroelectric energy at home in a way that has not been utilized before; it reflects the prospective possibility of new and renewable energy generation in the future.

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B. INVENTION SUMMARY

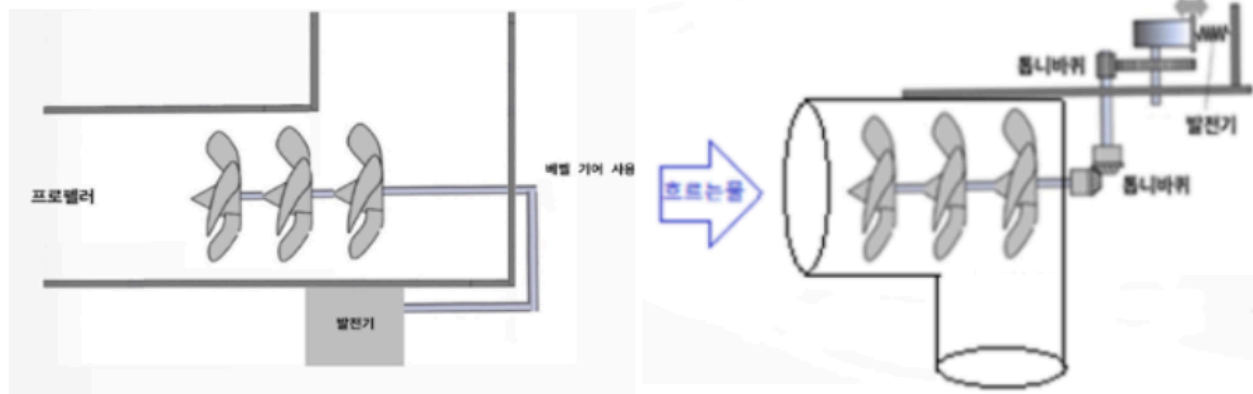
To create a hydroelectric generator between a shower head and a shower pipe, there has to be a piece of generator equipment to produce electricity using water's movement. Due to the energy from its kinetic energy, it will therefore spin the propellers inside the module and the gear attached to the axis-shaft-of-the propellers. Also, with the attachment of a gear and a shaft to the generator, the gear that will be turned by the propellers will work together with the latter gear. Finally, the generator will spin, creating energy. While the module is fixed, we propose an idea to minimize the pressure difference between the shower pipe and the space in the module. Overall, utilizing the strength of flowing water, the generator will bring out maximized electricity.



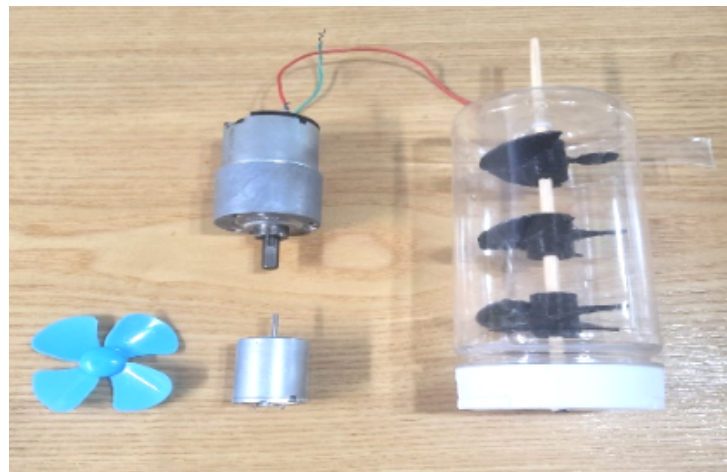
[Figure 6] Place of Installation for the Hydroelectric Module [Figure 7] Place of Installation (Front View)

C. DESIGN PROCESS

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[Figure 8] Concept Build

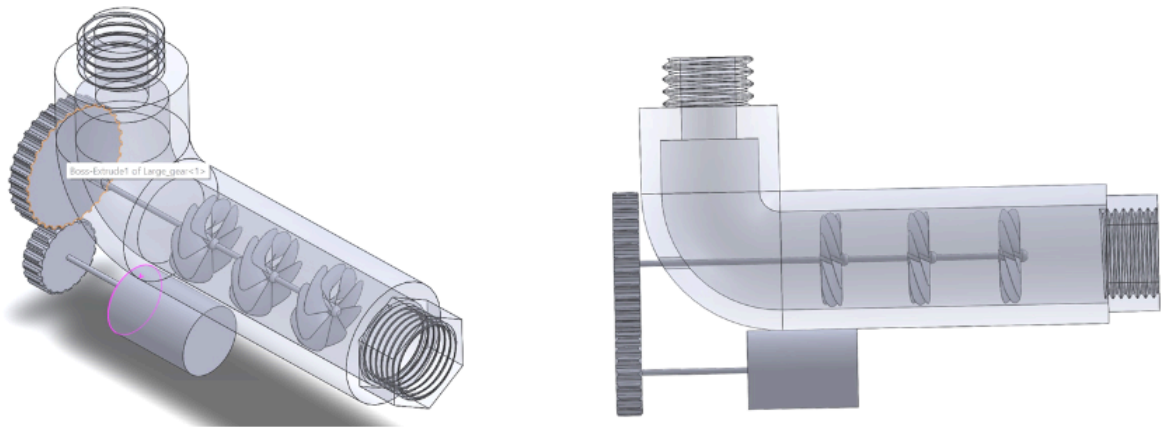


[Figure 9] Plastic Container Prototype

At first, our team drafted the first blueprint for the hydroelectric generator. In the drawings, we have struggled to implement this module in the pipeline of the shower where water pressure was sufficiently powerful to turn the propellers: thus creating a large amount of energy. The installation of this diminutive module, however, required taking down the wall and attaching the module to the pipeline using convoluted hardware. Furthermore, the bevel gears—gears that are turned at a perpendicular angle—were not only extremely complicated to stabilize the module but, therefore, very challenging to sustain its original condition for efficiency. Deeming it too difficult to meet the concept blueprints’ design, our team decided to seek alternatives.

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Instead of a 90-degree pipeline, we thought of making a straight tunnel for the propellers. In a plastic container with a bottom face with a small hole, we cut the hole into a size in which the propeller is capable of a smooth fit. As an experiment, our team put this prototype into use by means of spraying water; it first spun quickly when the water hit the propellers at a slant angle. However, when water was faced from the top, there was no movement; the plastic container including the propellers was inundated with water. This model allowed some kinetic energy to transpire, yet fastening the gears and the generators was impossible. Due to the slippery surface of the container as well as the danger of electrical shock, the mechanical parts can not be put together. These inconveniences forced our team to create another model that can not only be able to affix the parts and fix the gears together. The only way to successfully make a hydroelectric generator was through 3d modeling and 3d printing. As a result, our team utilized an application called Solidworks to construct an improved, more energy-efficient design.

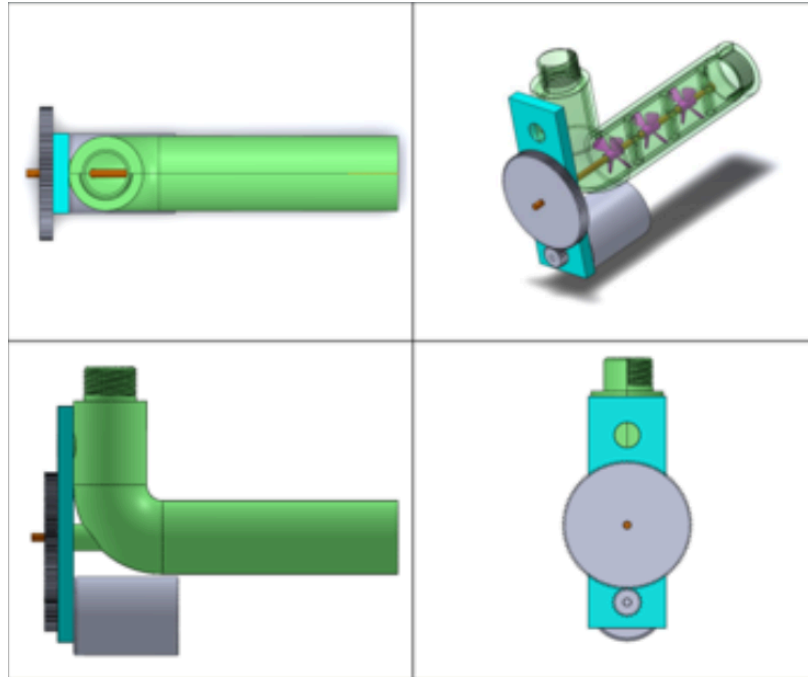


[Figure 10] SolidWorks 3d Design Blueprint, 1st Draft

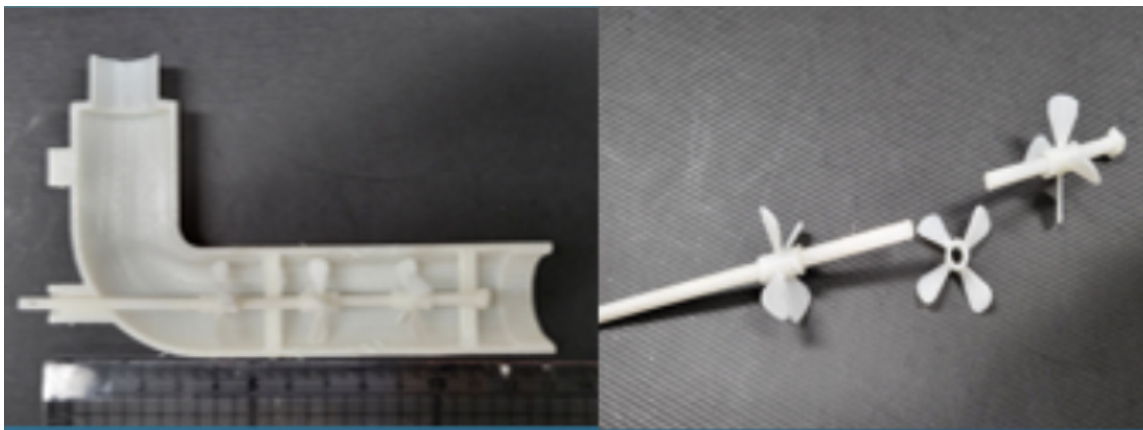
Through the 3d designing app, our team had better visualization of what our hydroelectric generator would appear like. As shown in the diagram, our team had finalized to have a 90-degree-angled tube with a propeller axis penetrated in the middle. With the axis and the gear fixed, they will turn the smaller gear attached to the generator. Specifically, the gear on the axis would be bigger so that the smaller gear with the electrical generator would revolve more: producing more electricity. Our team had resolved on placing the generator underneath the tube so that the gears would effectively turn. Despite this, the design lacked important dimensions as well as practicality: the gears did not fit well when assembled and the tube would be arduous to 3D print in its entirety. Finally, our team changed to spur gears for their simplicity and high efficiency. Our team had shown anxiety about the stability of the gears. If

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the gears spun too fast, there would be a possibility for the gears—thus the generator—to be damaged. These dilemmas once again pushed the team to build a new model—yet keep the shape from the first design.



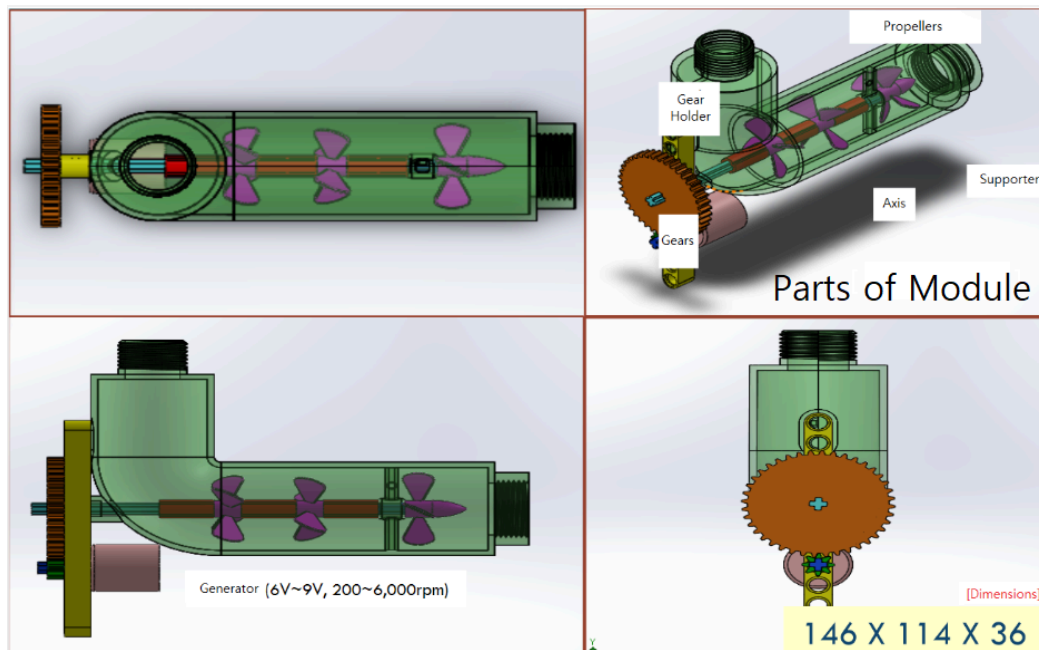
[Figure 11] SolidWorks 3d Design Blueprint, 2nd Draft



[Figure 12] 2nd Draft Design Physical Model

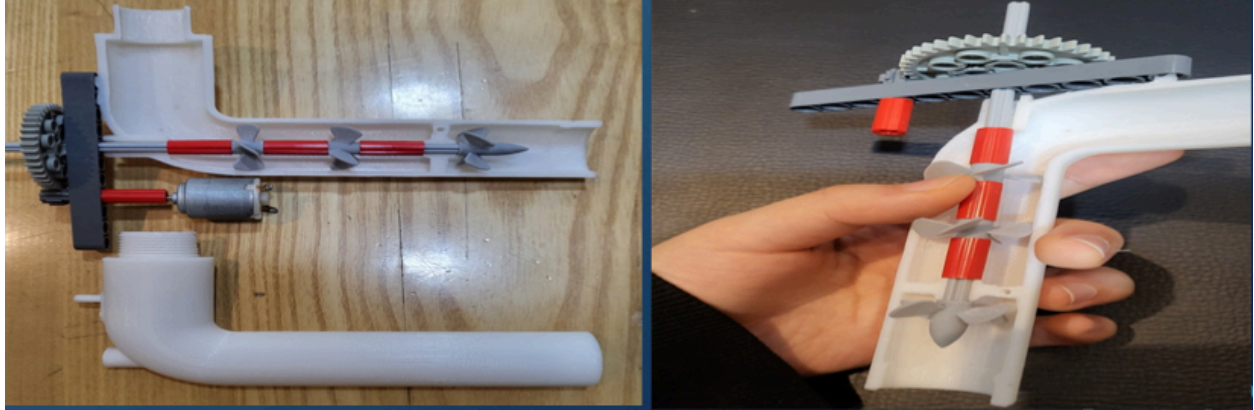
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For printing convenience, our team subsequently cut the tube into two halves. The altered design included a rectangular piece (with a small circular "holder") that would simultaneously hold the gears and the tube. Because we suggested putting this between the shower faucet and the shower head, we determined the precise dimensions for the shower faucet and the shower head's screw shape. The measurements on the entrance and exit holes were accurate, so the shower tubes fitted perfectly. Learning from the preceding design, a 0.3mm space—between the cylinder around the propeller axis— was made for the axis to rotate smoothly with minimum friction. In addition, there were small holes in the propellers in order to fix a pin inside to allow them to spin in one place. When we printed this structure using a 3d printer, the components were very fragile; all the small, thin pieces were bound to break. The propellers were fit for the axis, making another challenge to solve. Even though the whole model was more stabilized, the tube had to be rebuilt at a larger scale and the axis had to be substituted with another material that allows better motion.



[Figure 13] Finalized Design

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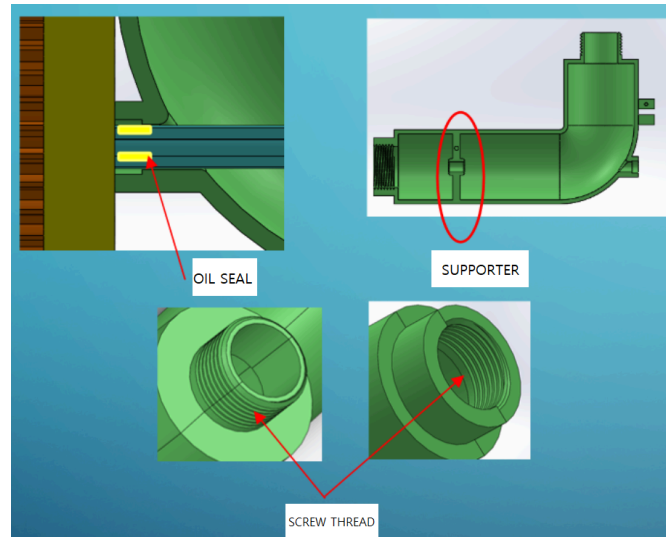


[Figure 14] Finalized Physical Model

Using most of the parts from the previous design, our team merely used LEGO for the propellers (with an arrowhead for the first propeller) and the gear holder. These two improvements supported the gears and fixed the axes of both propellers and the generator; lengthened pipe allows the propeller axis to spin without much resistance. Even though there was the use of Tetrix for better reinforcement, overall, there was better electricity generation compared to the previous model. In order to additionally save water, our team connected an ultrasonic sensor to a valve. This device would allow water to flow when the ultrasonic sensor receives a distance of less than 25 cm; when more than 25 cm, the valve would close and block the flow of water. Through an LCD screen, it would provide live data of the electricity(volts) generated from the mechanical device and the ultrasonic sensor to calculate distance. Finally, this design projected the most amount of electricity efficiently, becoming the final model of the micro-hydroelectric generator shower module.

D. SPECIFIC DESIGN CHARACTERISTICS

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[Figure 15] Final Model Characteristics

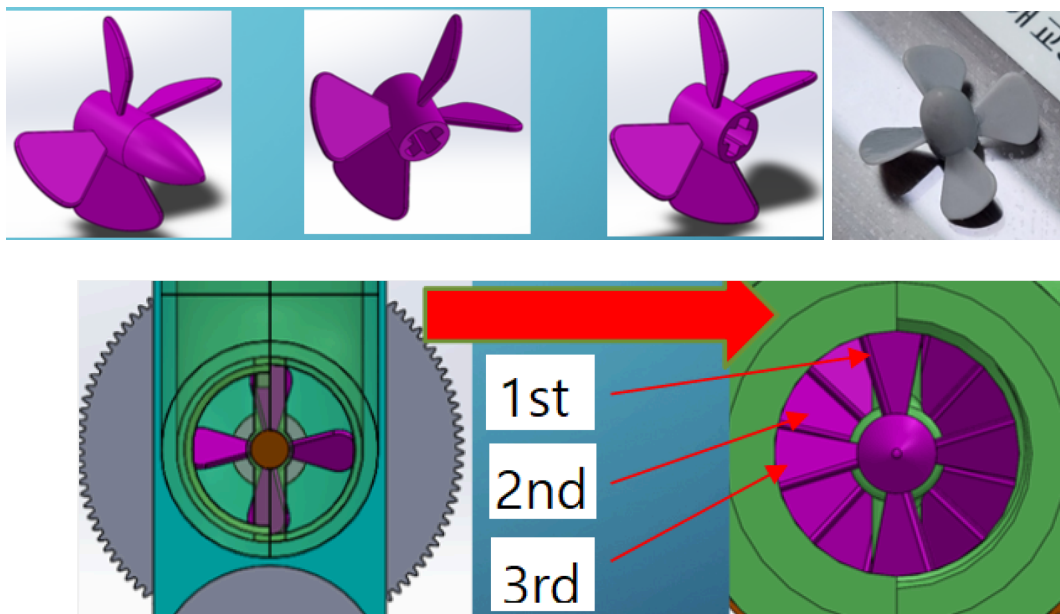


[Figure 16] Oil Seal

Distinctively from the previous designs, this model specifically greatly improved not only efficiency and stability but also the prevention of leakages. In the tube, a supporter was put in to hold the axis of the propellers. This addition would help the propellers to turn without moving freely; there may be a little more interruption in the flow of water, but this was the only method to maintain a stable axis. Secondly, a sealing was implemented in the hole for the axis in case of leakages. In the first and second models, when the axis was turned, water frequently flooded out through the 0.3 mm space between the axis and its hole. In order to solve this problem, our team suggested placing an oil seal rather than a metal bearing in between. Compared to a bearing—which still allows some leakage—an oil seal is composed of rubber but allows flexible movements even when fastened. Especially when the

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axis is turned very fast due to the strong water pressure, the nose gasket—an inner boundary space in the oil seal—pushes the water in from leaking. The rubber in the inner space, where the axis contacts, has a self-lubricating property that makes the axis spin with minimum abrasion or damage. Thirdly, our team inscribed a screw head for both the entrance and the exit holes of the module for better attachment to the shower. Due to this other factor, it can be very comfortably taken off or on by the user merely by screwing on both the faucet and the shower head line. The module's female screw thread would be connected to the faucet, whereas the male screw head would be connected to the shower head line.



[Figure 17] Propellers of Module

On top of the new design of the module's tube, our team designed the propellers in a way to increase electricity production and suppress as much pressure loss from colliding against the propellers. First, our team had to decide on what type of propellers should be used. We have researched many possibilities of the hydro turbines and reduced them to six: Francis, Fixed pitch propeller, Turgo, Pelton, Kaplan, and Crossflow. These six were categorized into two subjects: reaction turbines and impulse turbines.

Reaction turbines—Fixed pitch, Francis, and Kaplan—are any propellers that derive power from pressure drop; in other words, these turbines are used when in a tube where they use the velocity and the “push” of the water.

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Since these have a structure in which water is supplied and the turbines are spun in the circumferential direction of the tube, they are easy to install. However, there are many limitations: it was investigated that there are problems such as a rapid decrease in the capacity of small hydropower generation when the flow rate decreases and critical damage when turbines can not push down the water due to floating. On the other hand, impulse turbines–Pelton, Turgo, and Crossflow–convert mechanical energy into hydroelectric by a water emitter or water jet hitting buckets mostly from an outside environment. They have a problem because the turbines are not entirely submerged in water; the water is supplied only in a part of the aberration. Only the kinetic energy of the water is converted, resulting in low power generation efficiency.

However, because the module will use the kinetic energy/pressure of water and gravity within a tube, our team's resolution is on reaction turbines. Specifically, fixed-pitch propellers were the best candidate for the hydroelectric generator due to their decent electrical production and specialized ease of attachment. The fixed-pitch propellers are less expensive and take a simple operation but claim an efficiency range calculated through:

$$E = \frac{P_t}{P_w} \cdot 100\%$$

E = Efficiency of water turbine (%)

P_t = turbine power output (kW)

P_w = power of unrestrained water current(m³/sec)

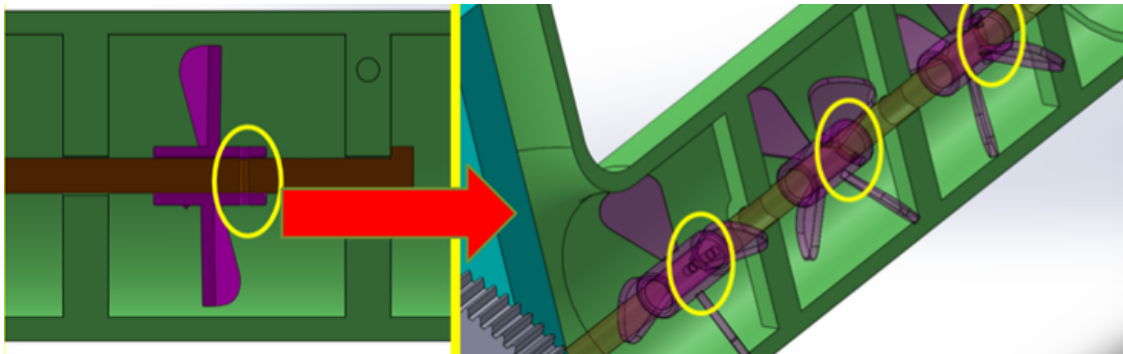
Thus, through this equation, the efficiency of the fixed-pitch propeller is: as high as 80% to 90% efficiency. Because fixed-pitch propellers are not only the most inexpensive compared to other reaction turbines but also produce a great amount of efficiency, fixed-pitch propellers were used.

Especially when positioning the propellers, our team considered the propellers to be placed at different angles rather than overlapping angles to maximize the surface area of the blades contacting water. By increasing the impacted area, it allows more rotations to occur; when the first blade hits the water and bounces off, the successive blade would be hit more with that pressure, then the third propeller would act correspondingly. Therefore, the propellers' blades were placed at a 30-degree angle to fill as much area of the cylindrical tube's cross-section.

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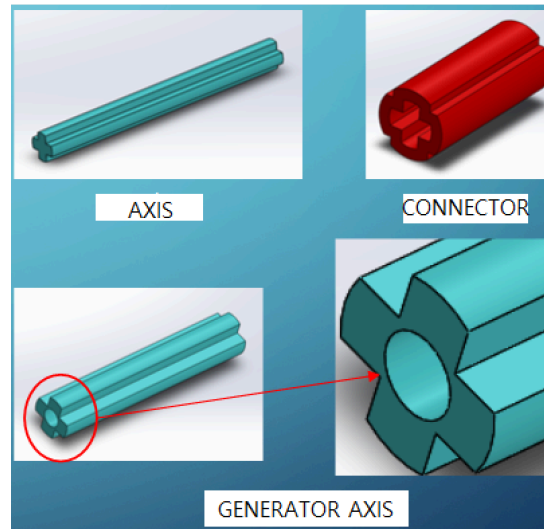
Another alteration for the propellers was the addition of a bullet-shaped cone for lessening aerodynamic loss in original water pressure. Based on Coanda Effect in fluid mechanics, in which an object maintains its inertial motion along a curvature but restricts flow against a solid surface, water will be penetrated exactly though rather than spreading and separating within the tube.

Thirdly, the module's gears were LEGO spur gears rather than the gears our team planned to use in the concept build. Our team thought it would be very demanding to stabilize a 90-degree angle; it would be very expensive to buy and keep it in its peak condition. As a result, our team finalized using spur gears, which are not only simple but still hold a high-efficiency percentage. In the final design of the module, the gears are used in a ratio of 2:1. It was originally thought that the bigger the ratio the greater the amount of production; however, from the numerous tests, our team found that the 2:1 ratio(40 teeth to 20 teeth) was the most efficient. Another reason why our gears with LEGO are cross-hole: by finding a piece of a cross axle that fits perfectly, there is no worry of extensively gluing the gears and their axes. Due to this simple solution, the gears and the axes were able to spin at the same time with almost no resistance.



[Figure 18] Pin Holes to Connectors Replacement

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[Figure 19] Axis/Gear Specifications

Fourthly, the axes and the connectors were uniquely used in the final design. In the second design, our team made pin holes in the propellers to hold them in one place when the pins were inserted. However, the 3D printer was not able to clearly show the extremely small holes. As a result, our team used lego cross axles and connectors as replacements. The longer axis is not just a single piece; the connectors attach the small axle parts on one end with the propellers on the other end. Due to the sturdy connectors, the propeller axis was impossible to be deformed or disassembled; it fits very well with the 40 teeth gear as well. The second shorter axis has a slight difference: it has a small hole drilled through. The small hole would hold the generator's axis tightly and restrain it from falling off from the module (when detached, it will not generate electricity).

E. FINAL MODEL AND TEST

a) Propeller Spin Test

We decided to test this prototype to determine if any additional changes need to be made after making the final model using 3D printing and LEGO parts. The first experiment was done with a real shower head and pipe. Due to the screw threads made on the tube, our team was successful in connecting the hydroelectric device to the shower faucet. Because this prior test was proposed to check if the propellers spun without any distraction or if the tube malfunctioned, any electrical devices—such as the generator—were not placed. After turning on the faucet, the pressure of the water lets the propellers in the module spin very quickly together with the gears. The two separate

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gears for the propellers and generator showed very similar motion to the propellers. The gears moved simultaneously with the propellers without any discerning friction or instability.

Confirmation was needed to corroborate if this shower module had, in actuality, been effectively designed. Three aspects that had to be checked were: the angle of collision, axis stability, and water pressure decrease. Several times, our team held the module at different angles to establish that no matter where the flowing water hit, it would still perform similarly in other cases. When tested, the gear spun very quickly in all positions, even though there were some locations where the gear turns faster. At the same time, our team learned that the propellers reached the highest speed when they were slanted at 75 degrees. When at 30 or 45, the gear's movements constantly paused and continued.

Secondly, before and after each experiment, our team scrutinized the axis of the propellers. In all of the situations, the axis seemed tightly fixed to spin at a uniform rather than in an arbitrary way (in the case of a loosened axis). Lastly, observing the water pressure in the entrance and the exit, our team visually judged that there was very little decrease in the power of the flowing water. Our team furthermore tested a smaller and bigger tube for pressure change. In a smaller tube, the narrow space caused water to collide with each other, slowing down its velocity, whereas the bigger tube let very little reduction.

< Speed of Flowing Water Calculation: 3 Steps >

- 1) Calculation of Volumetric Flow Rate

$$Q = v \cdot A$$

$$Q = \text{Volumetric Flow Rate (m}^3/\text{s)}$$

$$v = \text{Specific Volume (m}^3/\text{kg)}$$

$$A = \text{Mass Flow Rate (kg/s)}$$

- 2) Calculation of the Pipe Hole's Surface Area

$$S = \frac{r^2 \pi}{4}$$

$$S = \text{Surface Area of Pipe Hole (m}^2\text{)}$$

$$r = \text{Inner Pipehole Radius (mm)}$$

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3) Pipe Flow Rate

$$P = \frac{Q}{S}$$

P = Pipe Flow Rate (m/s)

Q = Volumetric Flow Rate (m³/s)

S = Surface Area of Pipe Hole (m²)

In summary, the device performed very well with 75-degree slanted propellers and a large-caliber tube brought more efficiency.

b) Gear Ratio Experiment, Voltage Experiment

To find the ratio that produces the most amount of electricity in a given time, our team experimented on five ratios, gear teeth of the gear of the propeller axis to those of the gear of the generator axis, respectfully: 5:1, 5:3, 2:1, 3:1, and 1:1. For the experiment, the generator was added on its axis. The generator's positive wire was plugged into the analog input of the Arduino, but the negative wire was then plugged into the GND pin. This system would allow the tester to visualize the amount of electricity produced by the generator when its axis is turned. However, when the generator provides the raw data to the Arduino board, it will not be a number; as a result, an equation must be used to convert it:

$$V = D \cdot \frac{Max.V}{Analog.S}$$

V = Total Voltage (volts)

D = Raw Data from generator

$Max. V$ = Maximum voltage of generator (volts)

$Analog.S$ = Analog signal limit (1023)

The calculated voltage will be shown through an LCD screen; it will display the average of all the live voltage values created from the generator.

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```
1 void setup() {  
2   // initialize serial communication at 9600 bits per second:  
3   Serial.begin(9600);  
4 }  
5  
6 // the loop routine runs over and over again forever:  
7 void loop() {  
8   // read the input on analog pin 0:  
9   int sensorValue = analogRead(A0);  
10  // Convert the analog reading (which goes from 0 - 1023) to a voltage (0 - 5V):  
11  float voltage = sensorValue * (5.0 / 1023.0);  
12  // print out the value you read:  
13  Serial.println(voltage);  
14 }  
15
```

[Figure 20] LCD Voltage Live Data Arduino Code

Unfortunately, using fluids to turn a generator required great strength; however, the axes turned merely by minute propellers were incapable of a heavy electrical generator. As a replacement, the module was tested with a DC motor(to emulate the water situation, preventing electric shock) connected to the axis of the propellers. The DC motor was left on 15 rpm for each gear ratio; through the Serial Monitor in Arduino, the live data (in voltages) could be seen. The chunks of data were later on organized into a spreadsheet of 40 entities to prepare to plot a graph.

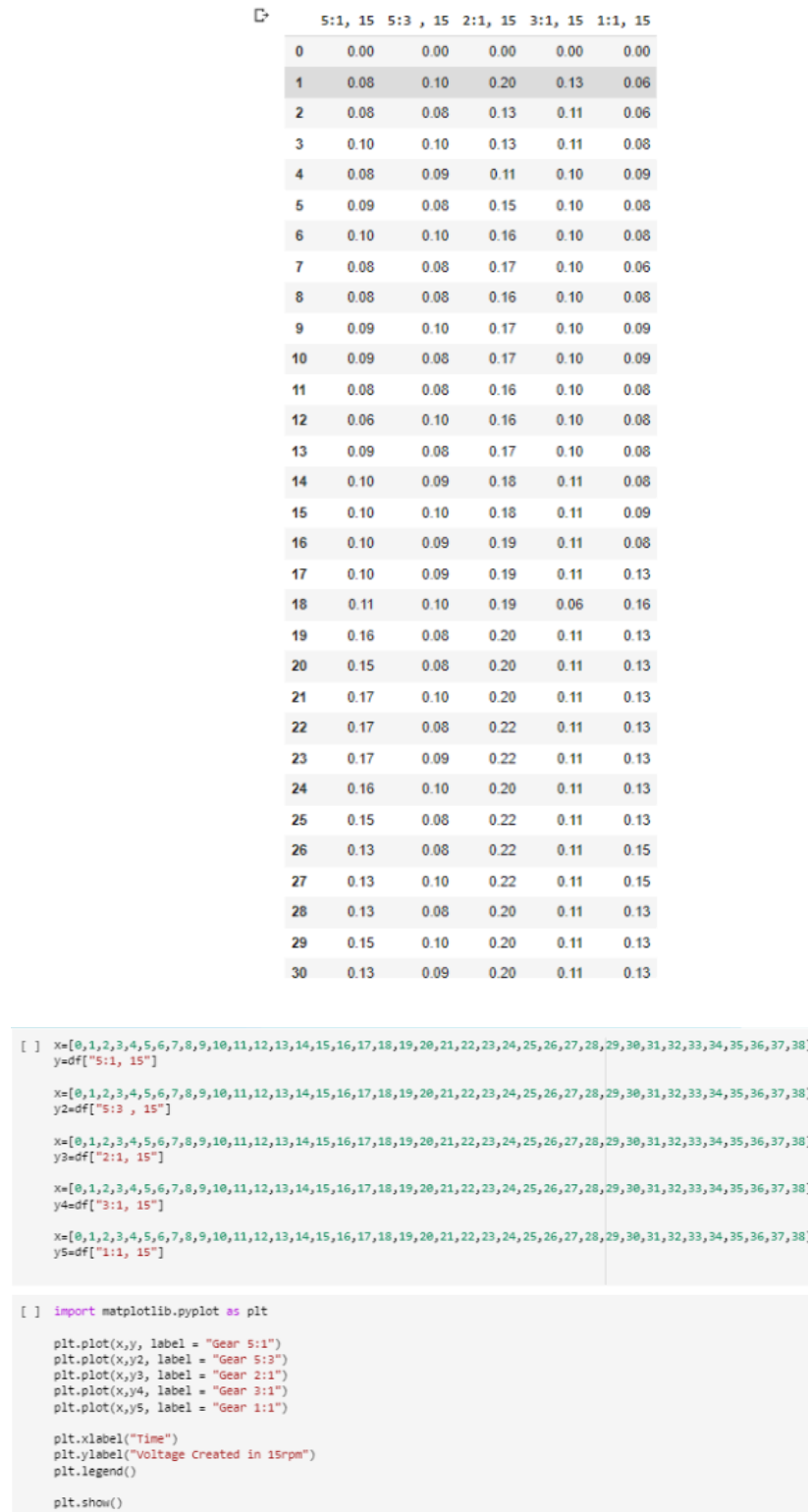
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	A	B	C	D	E
1	5.1, 15	5.3, 15	2.1, 15	3.1, 15	1.1, 15
2	0	0	0	0	0
3	0.08	0.1	0.2	0.13	0.06
4	0.08	0.08	0.13	0.11	0.06
5	0.1	0.1	0.13	0.11	0.08
6	0.08	0.09	0.11	0.1	0.09
7	0.09	0.08	0.15	0.1	0.08
8	0.1	0.1	0.16	0.1	0.08
9	0.08	0.08	0.17	0.1	0.06
10	0.08	0.08	0.16	0.1	0.08
11	0.09	0.1	0.17	0.1	0.09
12	0.09	0.08	0.17	0.1	0.09
13	0.08	0.08	0.16	0.1	0.08
14	0.06	0.1	0.16	0.1	0.08
15	0.09	0.08	0.17	0.1	0.08
16	0.1	0.09	0.18	0.11	0.08
17	0.1	0.1	0.18	0.11	0.09
18	0.1	0.09	0.19	0.11	0.08
19	0.1	0.09	0.19	0.11	0.13
20	0.11	0.1	0.19	0.06	0.16
21	0.16	0.08	0.2	0.11	0.13
22	0.15	0.08	0.2	0.11	0.13
23	0.17	0.1	0.2	0.11	0.13
24	0.17	0.08	0.22	0.11	0.13
25	0.17	0.09	0.22	0.11	0.13
26	0.16	0.1	0.2	0.11	0.13
27	0.15	0.08	0.22	0.11	0.13
28	0.13	0.08	0.22	0.11	0.15
29	0.13	0.1	0.22	0.11	0.15
30	0.13	0.08	0.2	0.11	0.13
31	0.15	0.1	0.2	0.11	0.13
32	0.13	0.09	0.2	0.11	0.13
33	0.15	0.09	0.2	0.1	0.13
34	0.13	0.1	0.2	0.1	0.13
35	0.13	0.09	0.2	0.11	0.13
36	0.12	0.08	0.2	0.1	0.13
37	0.13	0.1	0.19	0.11	0.13
38	0.11	0.1	0.2	0.1	0.13
39	0.11	0.09	0.2	0.11	0.13
40	0.13	0.1	0.19	0.1	0.13

[Figure 21] Generator Voltage Data Spreadsheet

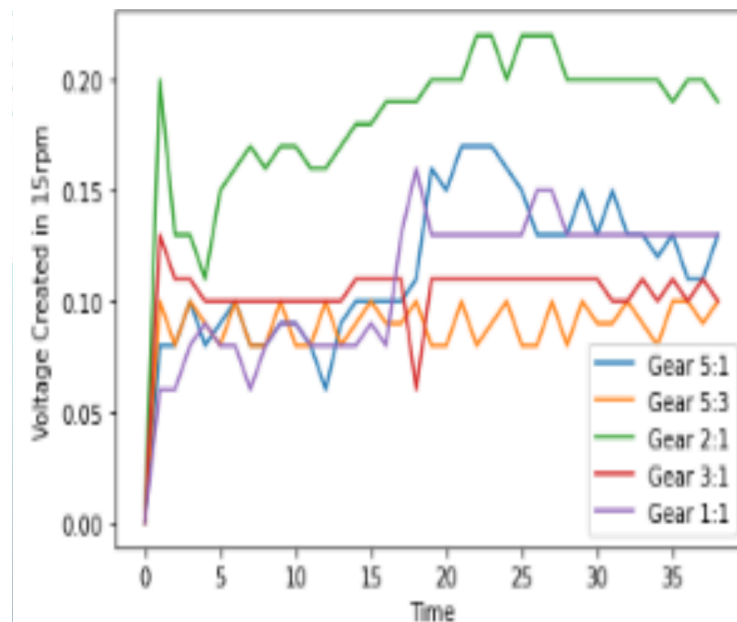
After condensing the data collected from the generator, a library(a specific function in programming) in Python called “Matplotlib” was utilized; the spreadsheet was turned into a “.csv” file for upload purposes. Google Colab, a Google engine based on Python, allowed the experiment data to be turned into a table(Dataframe) and a graph:

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[Figure 22] Google Colab Code and Dataframe

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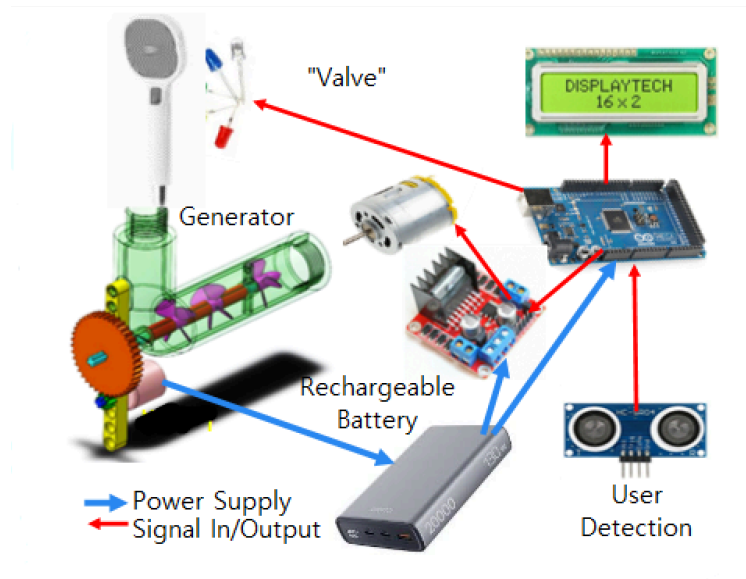


[Figure 23] Data Visual Graph

In the graph, there are five different types of graphs. Each is the voltage generated by the five distinct ratios. Eclipsing other statistics, the values of the 2:1 ratio had significantly higher voltage generation of 0.25 V at the maximum; other graphs have not been able to exceed 0.2 V. Due to the intense fluctuation of the graphs, our team tested this method three times consecutively; yet, all three results had no or very subtle variances. All in all, when the propeller axis's gear to the generator axis's gear ratio is 2:1, respectively, the generator produces the most amount of electricity. This result means when the ratio is 2:1, there has been a balance between the power of the bigger gear and the frequency of the revolutions of the smaller gear. Our team initially assumed a 5:1 ratio would be the most efficient in creating electricity because the smaller gear(generator) would spin more times if the other gear was bigger. However, this conjecture was refuted: more revolutions do not mean better energy production. All in all, if the hydroelectric shower module implements a gear ratio of 2:1, it will most efficiently produce a significant amount of energy. Even though a number of 0.23 V may be small in a short time; nevertheless, when used in a lengthened period of time—such as an hour—the generator would be creating great amounts of electricity.

F. IOT BUILD AND CODE

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[Figure 24] Basic Summary of IoT System

From the electricity produced from the module, our team proposed to use IoT(Internet of Things) to control and see the live data using Bluetooth. The IoT system formed with the module uses four components, excluding itself: Arduino Board, LCD screen, Ultrasonic Sensor, and LED. The Arduino board receives the program and executes signals to other electrical devices; the LCD screen shows the live data from the generator; the ultrasonic sensor detects if there is a person in front of it; and the LED acts as a “valve” to indicate if “water” is flowing or not. One may criticize our team for replacing a shower valve with a LED. However, making the valve was ultimately difficult due to more complex wiring and designing. This process requires the measurement of water flow as well as an idea of how to restrain highly-pressurized water in a shower pipe. For fewer complications and to put more importance on the module than the IoT build, our team considered this detail as secondary.

When the module is inputted with running water, the propellers will spin, turning gears and the generator. (However, for experiments, our team used a motor to mimic the water’s role.) The electricity that is created will be accumulated into a battery to store. The battery will therefore supply the power for Arduino to start running; the Arduino board will then give signals to the ultrasonic sensor, the LCD screen, and the LED. While the motor and the generator are running, the ultrasonic sensor will measure the distance and send this data back to the Arduino board. If the ultrasonic sensor detects a person at a distance of 15 cm or less, the board will consequently give an “on” signal to the LED, turning it on. On the other hand, if the ultrasonic sensor gives data of more than 15 cm, the

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Arduino will give an “off” signal, turning the LED off. Between each of these processes, the ultrasonic sensor’s data and the generator’s voltage data will also diverge to the LCD screen. The screen will simultaneously show the distance and the amount of electricity produced. Furthermore, our team implemented MIT App Inventor and a Bluetooth module on the Arduino to not only turn on and off the shower but also show voltage generation and distance using a phone. Therefore, this IoT system with the hydroelectric generator will bring convenience to the users by showing live data and conserving water with the use of a “valve.”

```
const int mpin1 = 2;
const int mpin2 = 3;

void setup() {
  Serial.begin(38400);
  Serial1.begin(38400);
  pinMode(mpin1, OUTPUT);
  pinMode(mpin2, OUTPUT);
}

void loop() {
  int ch;

  if (Serial.available()) {
    ch = Serial.read();
    Serial1.write(ch);
    Serial.write(ch);
  }
  if (Serial1.available()) {
    Serial.write(Serial1.read());
    digitalWrite(mpin1, HIGH);
    digitalWrite(mpin2, LOW);
  }
  else {
    digitalWrite(mpin1, LOW);
    digitalWrite(mpin2, LOW);
  }
}
```

[Figure 25] Bluetooth Arduino Code

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```
1  const int mpin1 = 2;
2  const int mpin2 = 3;
3  const int ENA = 9;
4  #include <LiquidCrystal_I2C.h>
5  #include <Wire.h>
6  float x = 0;
7  int y = 1;
8  LiquidCrystal_I2C lcd(0x27, 16, 2);
9  const int trig = 13; // ultra sonic trigger(output)
10 const int echo = 12; // ultrasonic receiver(input)
11 int a = 0; //variable to control the timing for the motor
12 const int led = 50;
13
14 void setup() {
15     // put your setup code here, to run once:
16     pinMode(mpin1, OUTPUT);
17     pinMode(mpin2, OUTPUT);
18     pinMode(ENA, OUTPUT);
19     pinMode(trig, OUTPUT);
20     pinMode(echo, INPUT);
21     pinMode(led, OUTPUT);
22     Serial.begin(9600);
23     lcd.init();
24     lcd.backlight();
25 }
26
27
28 void loop() {
29     digitalWrite(trig, HIGH); //fire trigger
30     delayMicroseconds(10); //wait
31     digitalWrite(trig, LOW); //turn off trigger
32     float distance = pulseIn(echo, HIGH) * 340 / 2 / 10000; //cm conversion
33
34
35     if (distance < 70) { //when user is there
36         if (a%2==0) { //only allows when x is an even number
37             digitalWrite(led, HIGH); //right direction(clock-wise)
38             a= a+1; //x=1, 3, 5, 7, 9...
39         }
40     }
41     else if (distance > 70 && a%2==1) { // when x is an odd number
42         digitalWrite(led, LOW); //counter-clockwise
43         a= a+1; //x=2, 4, 6, 8, 10...
44     }
45
46     analogWrite(ENA, 255);
47     digitalWrite(mpin1, HIGH); //right direction(clock-wise)
48     digitalWrite(mpin2, LOW);
49
50     int voltage = analogRead(A1);
51     float tvolt = voltage * (9 / 1023.0);
52     Serial.println(tvolt);
53     x= x + tvolt;
54     float total = x/y;
55
56     lcd.clear();
57     lcd.setCursor(4, 0);
58     lcd.print(distance);
59     lcd.setCursor(8,0);
60     lcd.print("cm");
61     lcd.setCursor(4, 1);
62     lcd.print(total);
63     lcd.setCursor(8, 1);
64     lcd.print("V");
65     delay(200);
66     y=y+1;
67
68     delay(300);
69
70 }
```

[Figure 26] IoT Full Code

III. CONCLUSION

A. HYPOTHESIS SUPPORT/REJECT

Overall, the hydroelectric module has created a maximum of 0.25 V—meaning it has fulfilled one of our hypotheses: one for creating electricity using the movement of water. Even though our team has understood that running water can produce electricity, our team’s hydroelectric generator has not reached our expectations of a maximum of 5 V. Nevertheless, our team strongly believes there is more possibility of improving after learning the mistakes and flaws of the module. We have brought bigger expectations to further develop this generator device so that it can produce more electricity efficiently and to build it more professionally to be able to release this as a patent or a product. Implementing a more enhanced “IoT-Implemented Environmental Hydroelectric Shower Module,” the module will bring great amounts of electricity from every household in South Korea. Even though an improved version may achieve producing more than 5 V may seem very diminutive at a smaller scale, if everyone uses this, it will produce imposing amounts of electricity, put down electrical bills together with the use of solar panels, and help the environment preserve more water.

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B. THIS INVENTION'S ORIGINALITY

The invention's specialized characteristics are laid out in the following table:

CHARACTERISTICS	EXPLANATION
1) Design of the propellers that makes the rotation the fastest in the shower faucet	Using SolidWorks, we made several designs of the propellers using 3D Printers and LEGO to find the complement angle. Finally, after experiments, LEGO was used due to its stability and efficiency in turning when water was sprayed. Our team found out, through experiments, that there was maximum efficiency when the blades were put at a 75-degree angle.
2) Pipe (tube) design to sustain flow rate, water suction power, and rotational power	Using SolidWorks and 3D Printers, our team made a sturdy tube that lets very minimal change in pressure when attached to a shower faucet. Due to this design and a supporter within it, the propellers were able to turn very quickly with less friction. Furthermore, by designing screw threads, the tube can be easily de/attached to a shower faucet.
3) Oil Seal	Rather than bearings, our team used an oil seal to stop the leakage of water when water was flowing into the module. An oil seal has better leakage prevention than a steel bearing but allows flexible movement of the axis.
4) Type of Propeller	Through the research of the types of hydraulic propellers, our team concluded the best and most effective type of propeller was a fixed pitch propeller. They were not only inexpensive but also easy to attach. They work very well in water and have a high percentage of efficiency.
5) Ratio and Type of Gears	With experiments of many ratios for gears using Arduino and creating a graph, it was shown that a gear ratio of 2:1 generated the most amount of electricity. Our team initially planned to use bevel gears, but thinking that it would be very difficult to build, we used spur gears for being simpler and highly efficient.
6) Automatic Shower Valve	Arduino, L298 Motors, and an Ultrasonic Sensor allowed our team to build a system of a shower valve. When the ultrasonic detects a person in front of it, the Arduino turns the L298 clockwise to open the water flow. On the other hand, when there is no one in front of the sensor, the L298 motor turns counterclockwise to close.
7) IoT Build and LCD	A Bluetooth module allows the user to see the live data

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	using MIT App Inventor. Secondly, for the user to see if the generator is working or not, our team put an LCD screen next to the module.
8) Use of Battery	A rechargeable battery will store the electricity made from the module, allowing it to use in versatile ways but not limited to: powering the automatic shower valve, charging phones, etc.
9) Generator and Axis	A generator has a maximum production of 9 V(~6000 rpm). Through the spin of the axes, the generator will rotate simultaneously, creating electricity. Specifically, our team made a hole in the LEGO axis to fix the generator.
10) Gear Holder	A gear holder holds the gears and the axes of the propellers and the generator for stability. It restricts any loose movements that affect electrical generation. Also, it connects the two halves of the tube.
11) Idea of Micro-Hydroelectric Module	This module is an innovation in creating electricity. It allows people to create electricity in every place where water flows. It repudiates the conventional thought that hydroelectric generation is only possible in dams or reservoirs.
12) Renewable Energy and Environmentalism	By powering the automatic shower valve with the electricity produced from the generator, it does not use any additional electricity from an outside source. Everything works on the energy from the module. Through the use of the valve, users can conserve any wasted water. Secondly, this module uses only the kinetic energy of water into electrical energy—without the use of batteries, fossil fuel, and other nonrenewable energy.

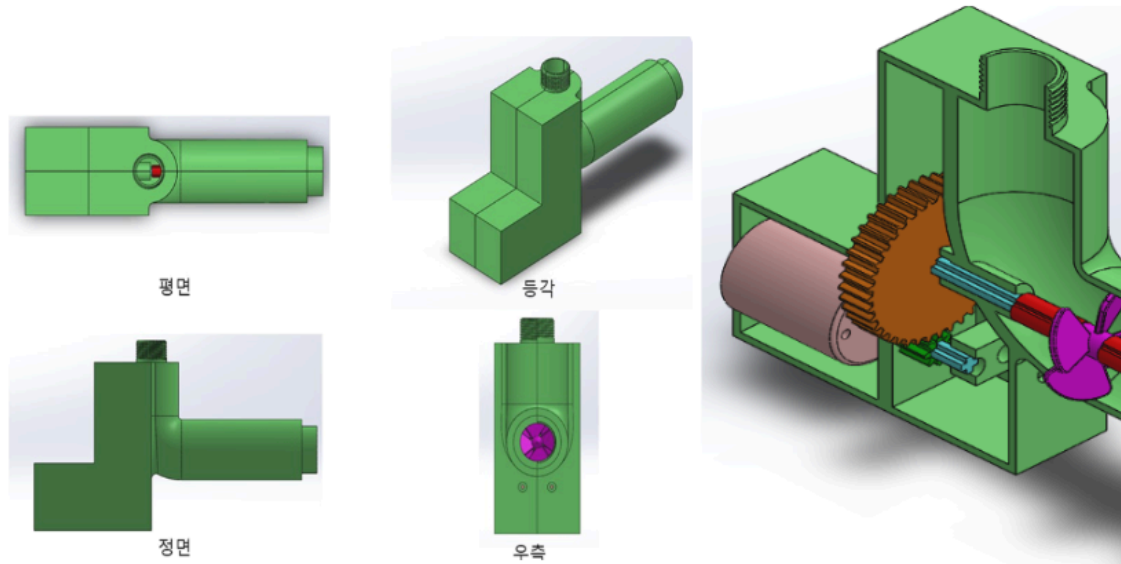
C. PROBLEMS IN TESTING, IMPROVEMENTS

During testing, there had been two problems we were not able to completely solve: the stability of the gears and the material of the module's tube.

First, even though our team had made a gear holder to keep the gears and the axes stable, it was not enough. Sometimes, the generator did not turn because of the high resistance due to the instability in the gears. Both the gears and the axes were not connected horizontally strongly enough. Due to this critical problem, the gears moved very strangely and loosely, jamming each other; the generator's axis quite frequently fell out of its hole. If these

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problems persist, the energy production will still remain very small and not be able to support the sensors and the motors used together with the module.

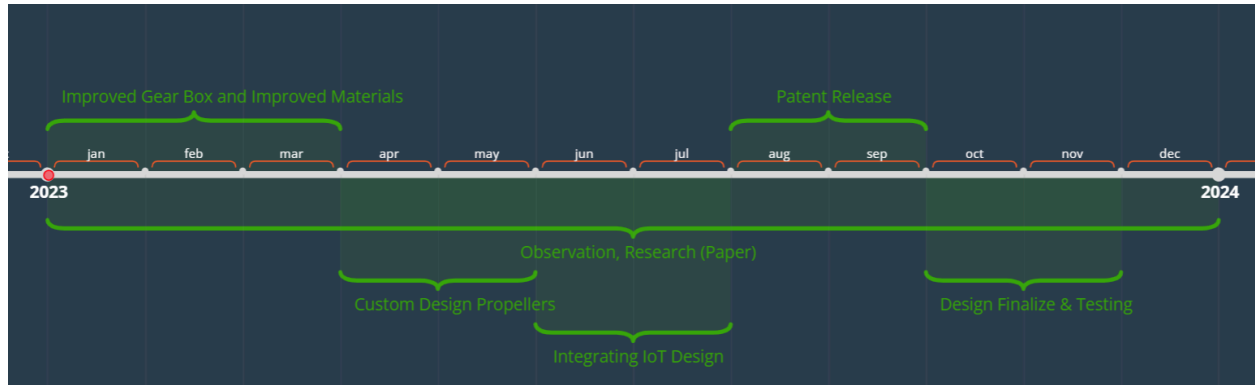


[Figure 27] Improved Gear Holder

As a result, our team decided to make a stronger gear holder that integrates the generator and the axes; however, the gears are separated from the main body. Differently from the previous design, the generator will not be put underneath the module but further away. The new design will also include more supporters to keep the gears/axes stable and fixed at one point; the generator will be supported from both sides of the wall to maintain its place. One very distinguishable change is the position of the axes: while the axis of the propellers faces away from the module, the generator's axis will face toward the module. This alteration will bring more space to the generator for connecting the battery and let the clip-like supporters on the walls hold up the two axes. From this integrated gear holder, the gears and the axes will no longer bring additional friction that prevents the generator from turning. Furthermore, to reduce more friction made by the abrasion of water against the pipe's rough plastic internal texture(3d-printed), it would be best to use smooth plastic or light metal. If these designs are applied, the module will create significantly more electricity and eliminate the danger of electric shock.

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D. “Schedule” for Improvements(How? When? Milestone—추후 계획, specific)



[Figure 28] Basic Timeline of Plans

In the future, our team promises to keep on working on designing and printing the prototypes so that we can continue finding better ideas. Right now, we are resolute in making an improved gearbox as shown above; however, the find will keep going. Starting from the new gearbox and using a different material for the tube, we plan to set four more goals to accomplish with each taking two to three months: designing a custom-designed set of propellers rather than LEGO, connecting the IoT parts integrally to hold it easier, making this idea as a patent, and finalizing the product. Our team wanted to create our customized components so that they will appear original when making a patent or releasing it in the market. However, nevertheless, we will consistently observe and research thoroughly while completing the goals one by one. Even though this process will not happen in a short period, we believe the way of producing energy through a micro-hydroelectric generator is a novel way to go against the overuse of electricity in our lives; thus, this module will greatly help in our everyday lives.

E. SUMMARY OF EFFECTS

1. CONSERVING WATER

Through the use of the module to create electricity to power up the sensors, the sensors allow an automatic valve that turns on and off when the user is there or not, respectively. It will stop the flow of water when it is unnecessary.

2. RENEWABLE ENERGY

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Electricity is created with the module just by the movement of water. From the kinetic energy of water, it spins the propellers—thus the generator. There are no other complications included but transfer from kinetic mechanical energy to electrical energy.

3. EASY TO ATTACH/DETACH

Because of the male/female screw threads, the user can simply screw this device onto the shower faucet. Then, after turning on the water, electricity will be produced. When there are any malfunctions, the user can unscrew the module from the faucet.

4. IoT-IMPLEMENTED

The micro-hydroelectric module uses LCD, Arduino, and Bluetooth to let the user know about the electricity generation and the distance measured by the ultrasonic sensor. With MIT App Inventor, one can see the live data on one's phone.

5. INEXPENSIVENESS

Purchasing the product will surely cost less than \$50. This module can last its decent structure from only one payment even after a long time. Even though there may rarely be some problems, it will not cost much. There is almost no maintenance fee.

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V. Feelings After Research (느낀 점)*****, *REFLECTION, THOUGHT*

While working on the project, it was very bittersweet. Even though it was fun to see it work and create electricity, it was sometimes very hard to fix the original module and improve it. All the time, I had to consider what went wrong and what alternatives there are to solve the problems. I also had to learn difficult subjects like SolidWorks, Arduino, IoT, and making the device! Furthermore, in the end, it was disappointing to see all my work crash down and only produce not even one volt of energy. I had a lot of expectations for this project, but seeing my final result, I felt defeated. However, whenever I take a shower or when I was writing this research paper, I had the time to organize my thoughts and see through what I have gone through just to make a miniature hydropower generator. I kept writing and contemplating every step I participated in with my team on this hard project. In the past, I felt sad; however, to see that I did so many things to come here, I feel—in actuality—some pride in myself and my team. Even though this was my first time making an invention together as a team for a long period, I

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couldn't believe I had touched hard things such as SolidWorks and Arduino. Furthermore, I felt amazing that I had a real copy of the invention working; an invention that can make electricity anywhere all around us. I was very excited to finally see the potential that energy could be found everywhere; if you just find a way to get hold of it, it can all turn into something we can use—just as water flowing from a shower can make electricity! Now, I feel more excited to gradually improve the module one by one so that they can one day produce electricity more efficiently. I hope that my finalized version in the future can be used by everyone. If the module makes a great amount of energy every day and accumulates, there would be mountains and mountains of electricity produced every year at a very inexpensive price(very small buying, and maintaining fee)! However, most importantly, I hope that this device can encourage people that environmental energy production is more tangible than how we conventionally think. I want people to have more interest in innovating the idea of renewable energy.

